*ACTIVITY 9*

# Evaluate Uncertainty and Product Benchmark Performance

ISO 14064 Internal Auditor Training
TGS-2025056357 · K6 · K7 · A5 · A6 · LO4 · 45 minutes

## Objective

Determine material uncertainty factors and compare the Lumina 400 with credible, method-aligned benchmarks.

## Scenario

The corrected Lumina 400 result is 186 kg CO2e per functional unit. Supplier aluminium data varies by ±25%, China grid factors differ by region, customer operating hours range widely and end-of-life recycling evidence is weak. Marketing compares the result with three competitor declarations: 175, 210 and 245 kg CO2e, but their functional units, lifetimes, boundaries and PCR editions differ.

## Materials

- Activity 8 model and contribution results
- Input ranges and data-quality ratings
- Three fictional competitor EPD extracts
- Uncertainty and benchmark workbook

## Activity questions

- Which uncertainty factors could change the decision?
- What sensitivity or scenario tests are proportionate?
- Are the competitor results comparable?
- What claim, if any, can HarbourLight support?

## Required evidence

- Uncertainty register
- Sensitivity and scenario results
- Benchmark comparability screen
- Performance and claim recommendation

## Detailed Procedure

**1.** Identify activity-data, factor, model, sampling, scenario and boundary-judgement uncertainty.

**2.** Rate likelihood, range, direction, evidence quality and decision impact for each factor.

**3.** Select high-impact weak inputs for sensitivity testing and document plausible ranges and sources.

**4.** Run one-at-a-time sensitivities and coherent scenarios for aluminium, grid, lifetime and recycling.

**5.** Determine whether ranges change hotspot ranking, total result, target status or proposed claim.

**6.** Screen each benchmark for functional unit, product performance, system boundary, allocation, geography, technology, GWP basis and vintage.

**7.** Exclude non-comparable records or normalise only where a defensible method and evidence exist.

**8.** Present ranges, comparability limitations and ranking without false precision.

**9.** Recommend approve, qualify, defer or reject the proposed product claim and state evidence needed for reconsideration.

## Acceptance Check

All material uncertainty classes are addressed; sensitivity links to decisions; only genuinely comparable benchmarks influence the conclusion; claim wording matches evidence and limitations.

## Learner completion checklist

| Done | Evidence / output | Reviewer comment |
|---|---|---|
| ■ | Uncertainty register | |
| ■ | Sensitivity and scenario results | |
| ■ | Benchmark comparability screen | |
| ■ | Performance and claim recommendation | |